ClearPath Deal Calculation Logic

Document: 01 of 10
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Overview

The ClearPath Analyzer accepts a property address or Zillow link and returns five core outputs:

OUTPUT	DESCRIPTION
ARV	After Repair Value — what the property is worth after improvements
Rehab Estimate	Estimated cost to repair/renovate based on condition
Rent Estimate	Estimated monthly rent if held as a rental
Cash Flow	Monthly net income if purchased as a rental
Flip Profit	Net profit if purchased, rehabbed, and resold

Inputs

VARIABLE	TYPE	SOURCE	NOTES
`address`	string	User input	Street address
`zillow_url`	string	User input	Optional alternative to address
`property_sqft`	number	API / Zillow	Square footage of property
`bedrooms`	number	API / Zillow	Bedroom count
`bathrooms`	number	API / Zillow	Bathroom count
`year_built`	number	API / Zillow	Used for rehab tier logic
`condition`	enum	User select	`cosmetic` / `light` / `medium` / `heavy` / `gut`
`purchase_price`	number	User input	What they plan to pay

`down_payment_pct`	number	User input	Default: 25%
`interest_rate`	number	System default	Default: 7.5%
`loan_term_years`	number	System default	Default: 30
`hold_months`	number	User input	Flip hold time, default: 6 months
`arv_comps`	array	API	List of recent sold prices in area (optional)

Processing Logic

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1. ARV (After Repair Value)

Method A — Comp-Based (preferred):

```
avg_price_per_sqft = SUM(comp.sold_price) / SUM(comp.sqft)  [from last 6 months, within 0.5 miles]
ARV = avg_price_per_sqft × property_sqft
```

Method B — Zestimate Adjustment (fallback):

```
ARV = zillow_zestimate × 1.10
```

> Note: Zestimates reflect current as-is value. A 10% premium approximates post-repair value for light-medium rehabs. Adjust multiplier by condition:
>
> - Cosmetic: × 1.05
> - Light: × 1.10
> - Medium: × 1.18
> - Heavy: × 1.25
> - Gut: × 1.35

Constraints:

- If ARV < $50,000 → flag as low-confidence
- If ARV > $1,000,000 → flag for manual review
- Round to nearest $1,000

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2. Rehab Estimate

Rehab Tiers (cost per square foot):

CONDITION	DESCRIPTION	COST/SQFT
`cosmetic`	Paint, carpet, fixtures	$10 – $20
`light`	Cosmetic + kitchen/bath updates	$20 – $35
`medium`	Light + roof, HVAC, systems	$35 – $55
`heavy`	Structural + full interior	$55 – $85
`gut`	Full demolition and rebuild	$85 – $150

Formula:

```
rehab_cost_low  = property_sqft × tier_low
rehab_cost_high = property_sqft × tier_high
rehab_estimate  = (rehab_cost_low + rehab_cost_high) / 2
```

Age Adjustment:

```
if year_built < 1970:
    rehab_estimate = rehab_estimate × 1.15   // older homes need more work
elif year_built < 1990:
    rehab_estimate = rehab_estimate × 1.07
```

Output: Display as a range — e.g., `$42,000 – $66,000` — not a single number.

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3. Rent Estimate

Method A — Rent Ratio (default):

```
monthly_rent = ARV × rent_to_value_ratio
```

Rent-to-Value Ratios by neighborhood tier (Chicago):

MARKET TIER	MONTHLY RATIO
Class A (North Side, Lincoln Park, Wicker Park)	0.6% – 0.8%
Class B (Logan Square, Pilsen, Bridgeport)	0.8% – 1.0%
Class C (South Shore, Englewood, Austin)	1.0% – 1.3%

> Default to 0.9% if market tier is unknown.

Formula:

```
monthly_rent = ARV × 0.009   // default
```

Method B — Bedroom-Based (fallback):

BEDROOMS	ESTIMATED MONTHLY RENT (CHICAGO AVG)
Studio	$1,100
1 BR	$1,400
2 BR	$1,800
3 BR	$2,200
4 BR	$2,700

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4. Cash Flow (Buy & Hold)

Step 1 — Loan Calculation:

```
loan_amount = purchase_price × (1 - down_payment_pct)

monthly_rate = interest_rate / 12
n = loan_term_years × 12

monthly_mortgage = loan_amount × (monthly_rate × (1 + monthly_rate)^n)
                                / ((1 + monthly_rate)^n - 1)
```

Step 2 — Monthly Expense Breakdown:

EXPENSE	RATE	NOTES
Vacancy	8% of rent	Assumes ~1 month/year vacant
Property Management	10% of rent	If using a PM company
Maintenance	6% of rent	Ongoing repairs
CapEx Reserve	5% of rent	Future big-ticket items
Insurance	$100/mo	Flat estimate (adjust by property)
Property Taxes	Varies	Pull from listing data or use 1.5% of ARV/12

```
effective_rent       = monthly_rent × (1 - 0.08)   // after vacancy
property_mgmt        = monthly_rent × 0.10
maintenance          = monthly_rent × 0.06
capex                = monthly_rent × 0.05
insurance            = 100
property_taxes       = (ARV × 0.015) / 12

total_expenses = property_mgmt + maintenance + capex + insurance + property_taxes
net_operating_income = effective_rent - total_expenses
monthly_cash_flow    = net_operating_income - monthly_mortgage
```

Step 3 — Annual Metrics:

```
annual_cash_flow       = monthly_cash_flow × 12
cash_on_cash_return    = annual_cash_flow / (purchase_price × down_payment_pct) × 100
```

Thresholds (display signals):

RESULT	SIGNAL
Cash flow > $300/mo	🟢 Strong deal
Cash flow $0–$300/mo	🟡 Marginal deal
Cash flow < $0/mo	🔴 Negative cash flow

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5. Flip Profit

Step 1 — Cost Stack:

```
purchase_price      = user input
rehab_cost          = rehab_estimate (midpoint)
closing_costs_buy   = purchase_price × 0.02      // title, inspection, etc.
closing_costs_sell  = ARV × 0.08                 // 6% agent + 2% closing
holding_costs       = (purchase_price + rehab_cost) × 0.01 × hold_months
                      // ~1% per month (taxes, insurance, utilities, loan interest)
```

Step 2 — Profit Calculation:

```
total_costs  = purchase_price + rehab_cost + closing_costs_buy
               + closing_costs_sell + holding_costs
flip_profit  = ARV - total_costs
ROI          = (flip_profit / total_costs) × 100
```

MAO (Maximum Allowable Offer):

```
MAO = (ARV × 0.70) - rehab_cost
```

> The 70% Rule — used to ensure a minimum profit margin is preserved.

Thresholds:

RESULT	SIGNAL
Profit > $30,000	🟢 Strong flip
Profit $10,000–$30,000	🟡 Thin margins
Profit < $10,000 or negative	🔴 Don't do it

Outputs

FIELD	FORMAT	EXAMPLE
ARV	`$XXX,XXX`	$185,000
Rehab Estimate	`$XX,XXX – $XX,XXX`	$28,000 – $44,000
Rent Estimate	`$X,XXX/mo`	$1,665/mo
Monthly Cash Flow	`+$XXX/mo` or `-$XXX/mo`	+$247/mo
Cash-on-Cash Return	`X.X%`	6.2%
Flip Profit	`$XX,XXX`	$34,200
MAO	`$XXX,XXX`	$154,500
Deal Signal	Emoji + label	🟢 Strong flip

Flow

```
User Input (address + condition + purchase price)
        ↓
Property Data Pull (Zillow API or manual)
        ↓
ARV Calculation
        ↓
```

```
Rehab Estimate
        ↓
Rent Estimate
        ↓
Cash Flow Calculation  ──→  Output: Monthly CF, CoC Return
        ↓
Flip Profit Calculation ──→  Output: Net Profit, MAO
        ↓
Deal Signal (Green / Yellow / Red)
        ↓
Display Results + Email Capture CTA
        ↓
If Chicago deal → Asset Group CTA
```

Edge Cases & Error Handling

SCENARIO	HANDLING
No comps available	Fall back to Zestimate method
ARV < purchase price	Flag: "Purchase price exceeds ARV — deal at risk"
Negative cash flow	Show result in red, display breakeven rent
flip_profit < 0	Show loss, display MAO to show right entry price
Missing sq footage	Prompt user to enter manually
Non-Chicago address	Suppress Asset Group CTA

Constants (System Defaults — Adjustable)

```
const DEFAULTS = {
  down_payment_pct: 0.25,
  interest_rate: 0.075,
  loan_term_years: 30,
  hold_months: 6,
  vacancy_rate: 0.08,
  mgmt_rate: 0.10,
  maintenance_rate: 0.06,
  capex_rate: 0.05,
  insurance_monthly: 100,
  property_tax_annual_rate: 0.015,
  closing_cost_buy_pct: 0.02,
  closing_cost_sell_pct: 0.08,
  holding_cost_monthly_pct: 0.01,
  mao_arv_multiplier: 0.70,
```

```
  default_rent_ratio: 0.009,
  zestimate_adjustment: {
    cosmetic: 1.05,
    light: 1.10,
    medium: 1.18,
    heavy: 1.25,
    gut: 1.35
  }
}
```

Next Document → `02_architecture.md` *— System Architecture*